

Ques. WHAT ARE THE DIFFERENT PULSE-WIDTH MODULATION (PWM) TECHNIQUES AND ALGORITHMS USED IN INVERTERS? WHICH TECHNIQUE IS BEST SUITED FOR PURE SINE-WAVE, SINGLE-PHASE INVERTERS?

A. Samiuddhin

Ans. PWM technique is used in such applications as power inverters, motor speed control and converters, among others. As technology advances, various PWM techniques have been developed for different industrial applications. There is no single PWM method that can suite all applications.

PWM technique used in power inverter applications is mainly for voltage control and reducing harmonic content in output voltage. In PWM inverters, width of output pulses is modulated to achieve the desired voltage control.

Different PWM techniques include single PWM, multiple PWM, sinusoidal PWM, modified sinusoidal PWM, space vector PWM, selective harmonic elimination and current-controlled PWM.

Sinusoidal PWM technique is commonly used in single-phase inverter applications. Generation of sinusoidal PWM signal is shown in Fig. 1. Gating signal can be generated by comparing a sinusoidal reference signal with a triangular carrier wave. Width of each pulse is changed proportionally to the amplitude of a sine wave evaluated at the centre of the same pulse.

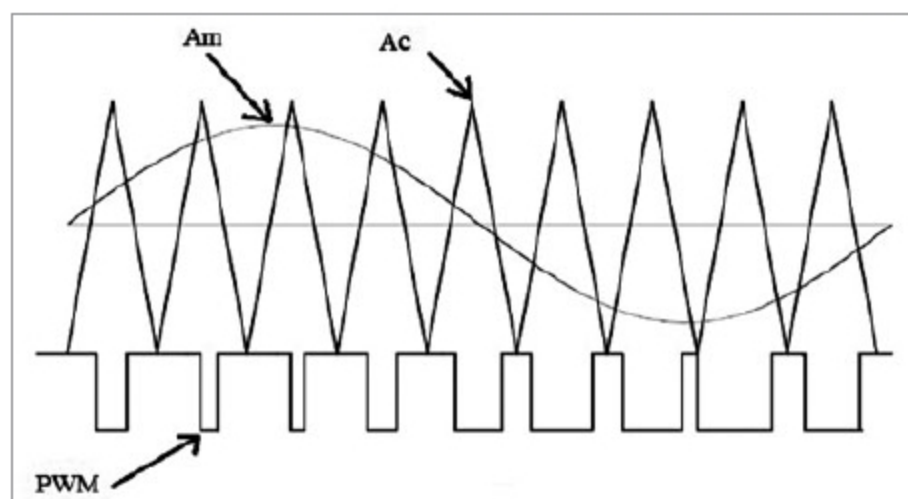


Fig. 1: Generation of sinusoidal PWM signal

When a modulating signal is a sinusoid of amplitude A_m , and amplitude of the triangular carrier is A_c , ratio $m = A_m/A_c$ is known as the modulation index. Output voltage can be controlled by modulation index m .

For details on PWM techniques and algorithms, you may refer to the following links:

- www.sciencedirect.com/science/article/pii/S1877705816342503
- [https://nptel.ac.in/courses/108105066/PDF/L-36\(DP\)\(PE\)%20\(\(EE\)NPTEL\).pdf](https://nptel.ac.in/courses/108105066/PDF/L-36(DP)(PE)%20((EE)NPTEL).pdf)
- www.theijes.com/papers/ICIEEE/F041051.pdf
- <http://shodhganga.inflibnet.ac.in/bitstream/10603/175704/3/chapter%202.pdf>

Q2. WHAT IS THE DIFFERENCE BETWEEN LED TV AND LCD TV?

Priyanka Shetti

A2. Both TVs use liquid crystal display (LCD) panels. The panels typically comprise two layers of polarising material with a liquid crystal solution in between. When electric current passes through the liquid, it causes the crystals to align, so that light can (or cannot) pass through. Basic LCD structure is shown in Fig. 2. Technically speaking, LED TVs are just LED-backlit LCD TVs.

LCD TVs use cold-cathode fluorescent lamps (CCFL) to provide backlighting. Whereas LED TVs use an array of smaller, more efficient LEDs to illuminate the screen.

The difference in lighting ele-

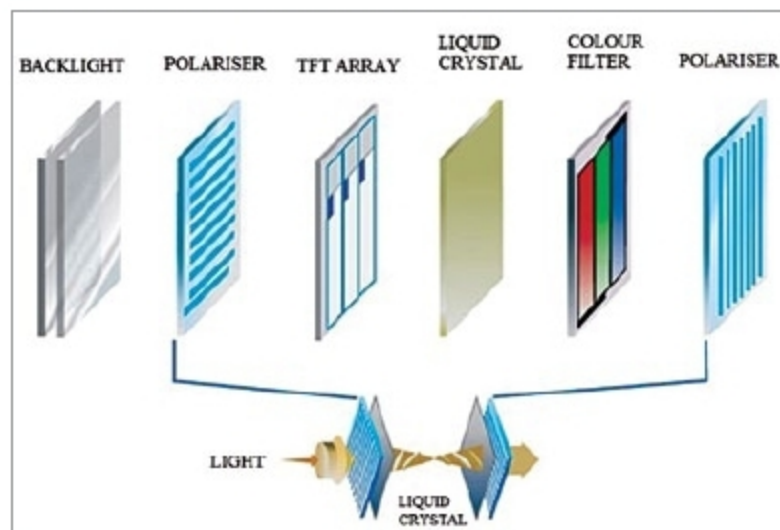


Fig. 2: Basic structure of LCD TV

LCD TV VERSUS LED TV

LCD TV	LED TV
Minimum thickness typically around 25.4mm to 50.8 mm (1-inch to 2-inch)	Minimum thickness less than 25.4mm (1-inch)
Cheaper	Costlier
Backlighting done using CCFLs	Backlighting done using LEDs
Consumes more power	Consumes less power

ments and component placement makes LED TVs thinner than LCD TVs. LED TVs run with greater energy efficiency and can provide a clearer, better picture than general LCD TVs.

LED TVs work with distinct RGB (red, green, blue) lights to produce more realistic and sharper colours. LED-dimming capability on backlighting in LED TVs allows the picture to display with a truer black colour and to block more light from passing through the panel. Since LCD TVs are thin-screened, these have angle-viewing and anti-glare issues.

Some differences between LCD TVs and LED TVs are given in the table above.

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